



Слова , произнесенные **Jacob Witten**

Выключатель для вредных генов

25 августа 2026 года



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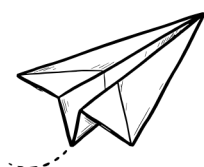
16 минут

Что, если вместо того, чтобы редактировать ген или блокировать вырабатываемый им белок, можно просто отключить его?

Если вам нужна метафора, описывающая биомедицинские исследования, представьте, что по лицу биолога десятилетиями бьют сапогом. На сапоге написано «медицинская химия». Это не должно восприниматься как оскорбление в адрес медицинских химиков, которые героически осваивают, пожалуй, самую сложную область биомедицины.¹ Я имею в виду вот что. У биолога может быть четкая гипотеза о том, как остановить развитие болезни, например: блокирование определенного белка снизит уровень холестерина в крови и уменьшит риск сердечных заболеваний. Вполне естественно, что биолог хотел бы проверить эту гипотезу. Но не существует волшебной кнопки, которая могла бы отключить этот белок; его должно блокировать какое-то физическое вещество. Обычно это небольшое химическое соединение, тщательно разработанное для того, чтобы его можно было ввести в организм, связать с белком, заблокировать его и больше ничего не делать. На практике все почти никогда не идет по плану, и это основная причина, по которой большинство лекарств терпят неудачу в клинических испытаниях.



Ситуация еще хуже: химики-фармацевты с трудом находят способы взаимодействия с большинством белков. Большинство белков человека считаются «непригодными для разработки лекарств», то есть ни одна из малых молекул, которые мы можем синтезировать в настоящее время, не может эффективно с ними связываться — либо потому, что у этих белков нет удобного «кармана» для лекарств, либо потому, что они «по своей природе неупорядоченные», то есть имеют слабую когерентную структуру.² В конечном итоге поиск новых лекарств сосредотачивается на тех мишенях, которые поддаются лекарственному воздействию, — как в притче о пьянице, который ищет ключи под уличным фонарем, потому что там светит свет.



Информационный бюллетень "Незавершенные работы"

Получайте новые статьи из рубрики Works in Progress на свой почтовый ящик.

Даже в этих случаях успех далеко не гарантирован. Препарат, который работает в лабораторных условиях, может разрушиться в организме до того, как достигнет белка-мишени, вступить во взаимодействие с другими белками и вызвать токсические побочные эффекты или не усвоиться и не распространиться по нужным тканям. Мы настолько плохо умеем предсказывать, как низкомолекулярные препараты будут действовать в организме, что даже если препарат и подействует, то не по той причине, которая предполагалась изначально!³



Но что, если бы вместо того, чтобы тратить годы на оптимизацию низкомолекулярных соединений для лечения всего одного заболевания, и при этом постоянно помнить о том, что ни один план не выдерживает первого контакта с противником, у нас действительно была бы волшебная кнопка, позволяющая отключить любой выбранный нами ген? В таком случае мы могли бы лечить не только повышенный уровень холестерина, но и десятки смертельных заболеваний, от болезни Альцгеймера до диабета и болезни Гентингтона.

Добро пожаловать в мир терапии малыми интерферирующими РНК.

Подавление экспрессии генов

История малых интерферирующих РНК начинается в 1990 году, когда ученые из корпорации DNA Plant Technology в Калифорнии пытались выяснить, почему их петунии стали белыми. На самом деле они пытались сделать петунии темнее, добавив дополнительную копию гена фермента, который вырабатывает пигмент. Предполагалось, что дополнительная копия гена фермента приведет к увеличению количества фермента и, следовательно, пигмента. Но каким-то образом дополнительная копия не стимулировала, а подавляла выработку фермента. Этот загадочный эффект называли «косупрессией».

Последующая работа других исследователей выявила его причину: особый вид молекул РНК, микроРНК, препятствовал выработке фермента. За это открытие Эндрю Файр и Крейг Мелло позже получили Нобелевскую премию.

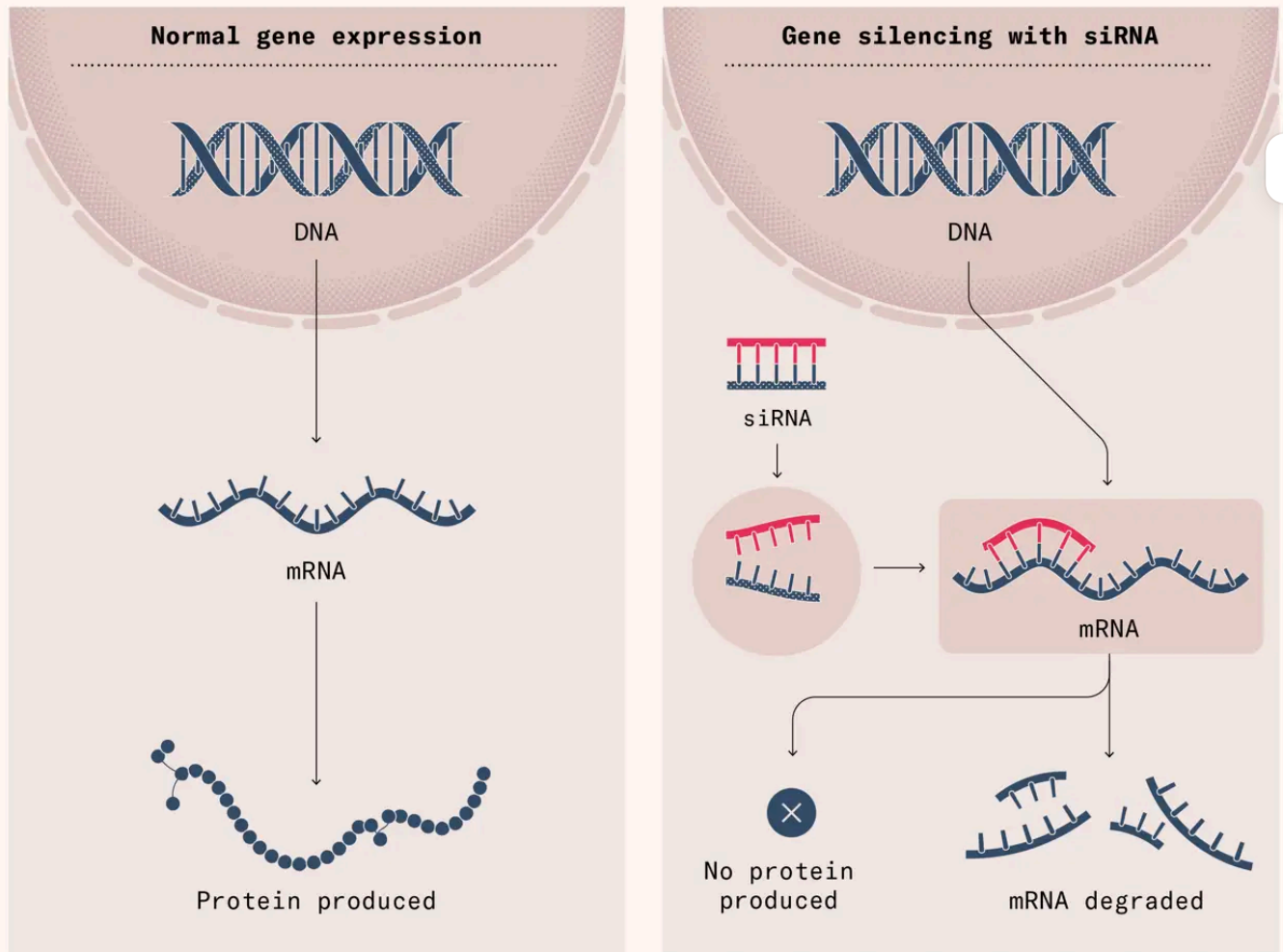
Biologists had already known about RNA for several decades. Similar to DNA, RNA is a complex molecule built from a chain of smaller building blocks that encode genetic information. But the two differ in one important detail: the building blocks of RNA have one extra oxygen atom compared to DNA (hence its name, ribonucleic acid, as opposed to DNA's 'deoxy'-ribonucleic acid). This oxygen is prone to chemical reactions that can break RNA molecules. DNA is stabler and is therefore used to store genetic information in all multicellular life. RNA molecules tend to be short-lived, and take many different forms within cells, carrying instructions, helping to assemble proteins, regulating genes, and catalyzing reactions.⁴ And while DNA is 'double-stranded', with two linked strands of building blocks twisting into its famous helix shape, RNA often exists as just one strand.

DNA contains the instructions for making proteins, from the enzymes that digest food to the keratin and collagen that form our hair and skin. But to get from DNA to protein, the genetic code is first transcribed into an intermediate RNA molecule, called 'messenger RNA' or mRNA, which is then translated into protein. If the mRNA is destroyed, the protein won't be built.

That's exactly what siRNA, or 'small interfering RNA', does: it's a short strand of RNA that interferes with the production of protein by destroying mRNA. An siRNA molecule binds to a piece of mRNA with a matching sequence, like a kind of barcode, and targets it for destruction by the cell's gene-silencing machinery, which cuts the mRNA into pieces.

siRNA silences genes by destroying their messenger molecules

Normally, DNA produces mRNA which builds proteins. siRNA binds to mRNA and degrades it, which blocks protein production.



siRNA also explains what happened in the petunias. Adding an extra copy of the pigment-producing gene triggered the petunias to develop siRNA targeting that sequence, and because the newly introduced gene and the original gene shared the same sequence, both were silenced, the flowers lost their pigment and turned white. This is a naturally occurring process: siRNA is used to 'silence' unwanted genes, such as those of viruses that have entered the cell. But we can also use it to artificially block the production of proteins that cause disease.

PARENTAL



Transgenote
218.11



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218.43



Transgenote
218.56



Transgenote
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Four petunia lines with pigment genes silenced by siRNA, each producing flowers with unpredictable patches of color loss.

IMAGE

Napoli et al. (1990)

siRNA therapy

For some diseases, gene silencing has already succeeded. One example is hereditary transthyretin amyloidosis, a condition caused by a mutation in the

transthyretin gene, which makes that protein misfold and clump together in the bloodstream, nerves, and heart. Over the years, the deposits build up, damaging nerves and stiffening the heart muscles, which can eventually lead to heart failure. Until recently, the condition meant a life expectancy roughly a decade shorter than average.⁵

In 2018, the FDA approved the very first RNA interference drug, called patisiran, to treat the nerve damage caused by this disease. A newer, longer-lasting version called vutrisiran was approved in 2022. Both drugs use siRNA to prevent the production of the misfolded protein so that clumps don't form, which halts or slows down progression of the disease.



This approach could be adapted to many different diseases, from high cholesterol to Alzheimer's, whose progression hinges on a small number of critical proteins. RNA interference makes it possible to target those proteins specifically and stop them from being produced.

Have a gene you'd like to silence? Identify the gene's sequence, synthesize a short siRNA sequence to match it, deliver the siRNA into the right cells, and bam – instant knockdown. It sounds simple enough in theory, but that last step, getting siRNA into the right cells, has proven surprisingly difficult. Only after decades of grinding organic chemistry has it been solved.

The delivery problem

If you just injected siRNA into a patient's veins, nothing useful would happen. There are at least three reasons for this. First, the body is full of enzymes that recognize and destroy unprotected RNA, including siRNA. Second, even a perfectly intact siRNA molecule would do nothing in the bloodstream, as it needs to get inside a cell to reach its target mRNA. Third, the way it would get into a cell involves being enveloped in a small bubble called an endosome, where it would be broken down by digestive enzymes. So an siRNA molecule would need to stay intact, enter its target cells, and then somehow escape from the endosome without being digested.

One way to solve these challenges is to use nature's own delivery system: viruses. Viruses have spent billions of years evolving to get past our cells' defences, find their way in, and release their proteins. So what if we used viruses to carry siRNA into our cells? Researchers have tried to do just that, by

engineering harmless viruses to carry and deliver siRNA into specific tissues, including the brain. But viruses come with a drawback: just as they have evolved to infiltrate our cells, our immune systems have spent hundreds of millions of years learning to recognize and destroy them. Even carefully designed viral delivery systems can trigger immune reactions, especially if they're used repeatedly, which can be risky for patients and make treatments less effective.

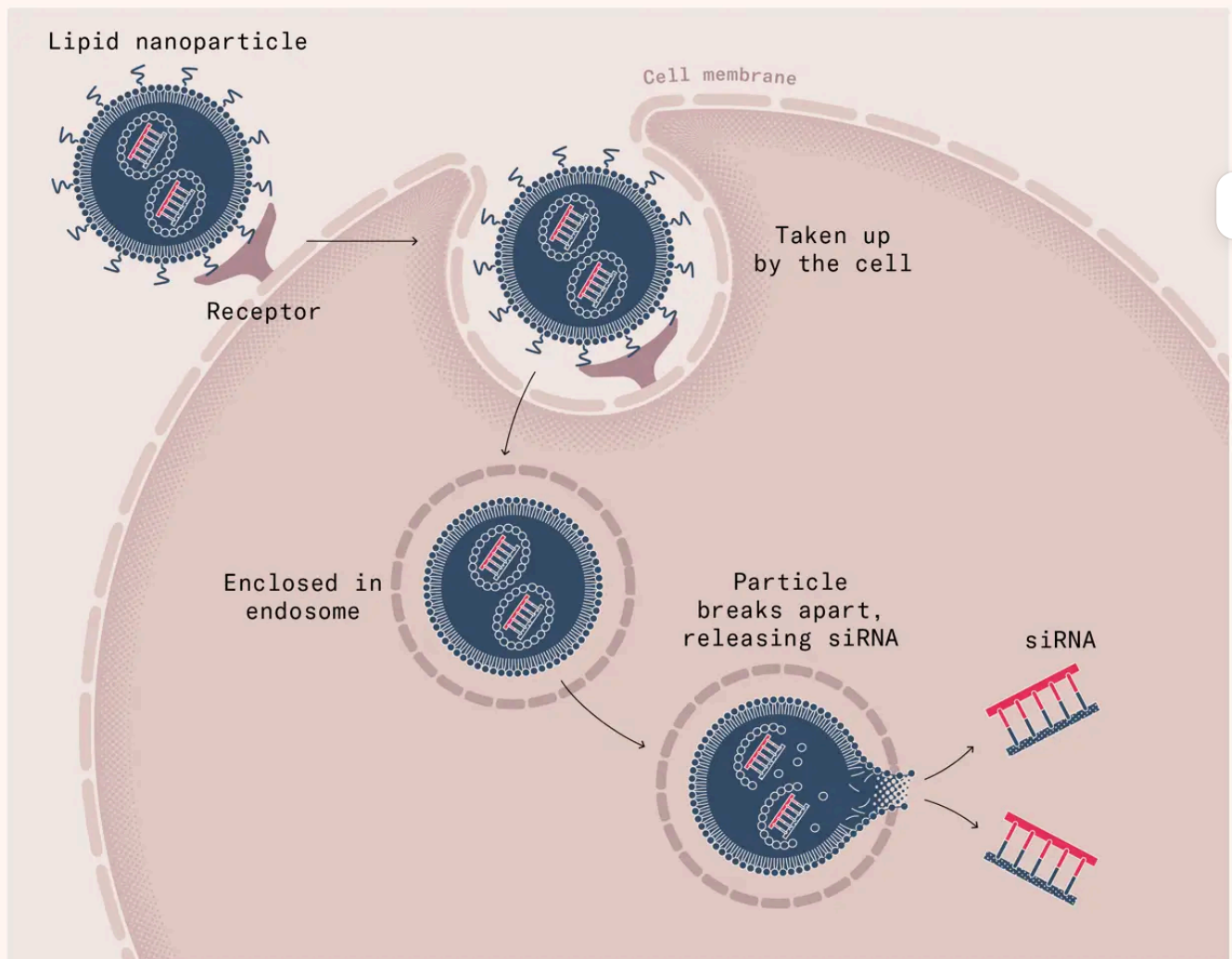


Signed, sealed, delivered

Another approach has focused on packaging RNA inside tiny fat-like particles, called liposomes, to protect and help it enter cells. Liposomes were first explored in the 1960s, and could shield RNA from being broken down in the body to some degree. But liposomes were far less successful at entering cells, because they bound only weakly to their outer membranes, and were even worse at escaping endosomes and releasing their payload of siRNA into the cells.

Lipid nanoparticles smuggle siRNA into cells

Lipid nanoparticles containing siRNA bind to a cell surface receptor, get swallowed into an endosome, then disassemble to release the siRNA into the cell.



Adapted from Alnylam

That challenge was overcome in the 1980s, when researchers tweaked liposomes to bind more tightly to negatively charged cell membranes, by incorporating lipids with a positive electric charge. While this helped get the RNA into cells, it came at a cost: the strong positive charge made them interact with cell membranes too strongly and damage them.

The next breakthrough was more subtle. Instead of using lipids that were always charged, researchers incorporated lipids that could change their charge depending on the acidity of their environment, called 'ionizable lipids'. These lipids are neutral in the bloodstream, minimizing damage, but turn positive once inside an endosome, as endosomes are acidic. This destabilizes the endosome's membrane, releasing the siRNA into the cell.

Even after coming up with that basic idea, the problem was nowhere near solved. It took more than a decade of tinkering with ionizable lipid chemistry, leading to a thousand-fold improvement in potency, to produce lipid nanoparticles that performed all these three steps of delivery (stability, uptake into cells, and endosomal escape) safely and efficiently.⁶

Patisiran, the first FDA-approved siRNA drug, was the first success that took this approach, by delivering siRNA in ionizable lipid nanoparticles to silence the mutated transthyretin gene in liver cells and slow the progression of amyloidosis. It was proof that siRNA could be turned into a real medicine. But it also arrived, ironically, just as the use of lipid nanoparticles for siRNA was being eclipsed by a more efficient method.⁷ What if siRNA didn't need a carrier at all?

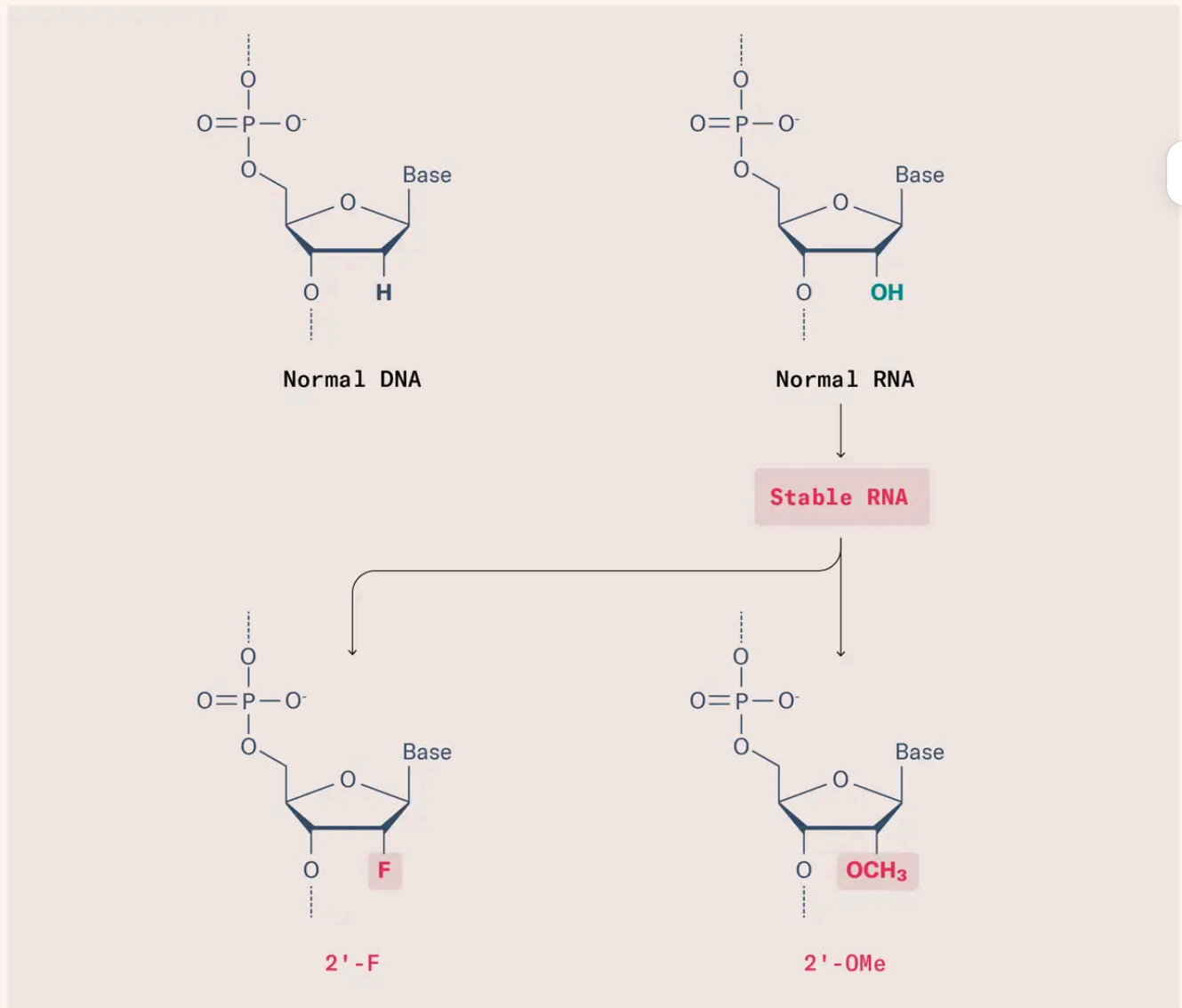
Naked siRNA

So far, researchers had focused on shielding siRNA from being broken down inside the body. But then they tried a new tack: making the siRNA molecule itself sturdier so that it could survive unclothed.

To understand how this was possible, let's return to the difference between DNA and RNA. RNA has an extra oxygen atom, which makes it vulnerable to reactions that break it apart. By chemically modifying that pesky oxygen – for example, by swapping the oxygen for less reactive fluorine or hydrogen, or linking a methyl group (a highly stable formulation of one carbon atom and three hydrogen atoms) to it – the RNA molecule can be made much more durable. These small, precise chemical changes allowed Alnylam Pharmaceuticals (which also developed patisiran) to create siRNA molecules with almost indefinite protection against degradation when injected naked.

Swapping one atom makes RNA more stable

The 2' position (bottom of the sugar ring) is RNA's weak point. Replacing the vulnerable -OH group with a fluorine (-F) or methyl (-OMe) makes the molecule much more stable in the body.



With stability solved, the next challenge was guiding this naked siRNA to its target. The siRNA drugs carried by lipid nanoparticles naturally accumulated in the liver, because lipid nanoparticles impersonate cholesterol which get processed there. This made the liver an easy target for siRNA drugs. But without lipid nanoparticles, naked siRNA molecules wouldn't concentrate at a target organ. So researchers attached a sort of address label to get them there: three copies of a small sugar molecule called GalNAc. Liver cells have receptors for GalNAc on their surfaces, which recognize the sugar and bring it inside their endosomes. This directed siRNA to its intended destination.

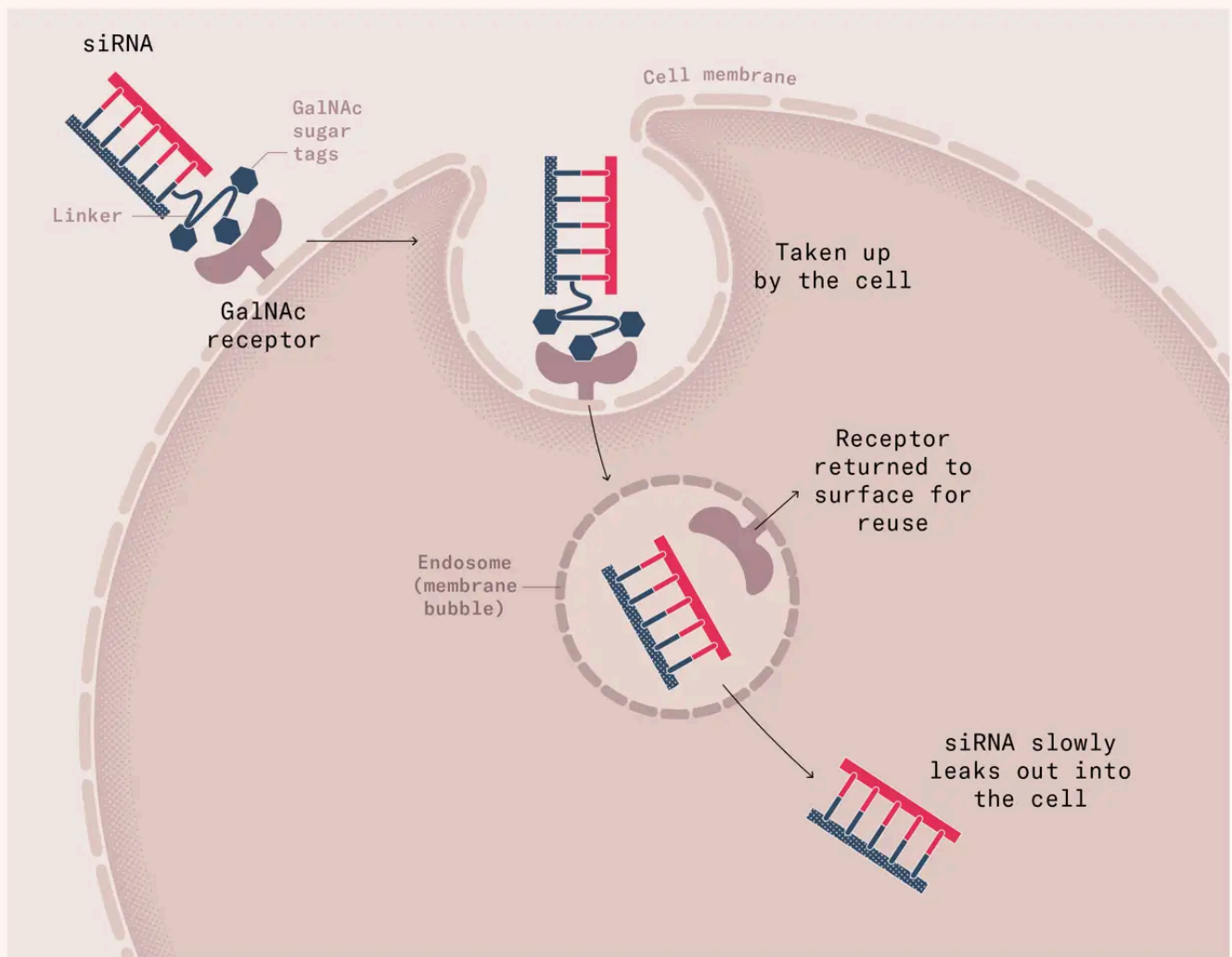
Alnylam had cracked the first two hurdles of delivering naked siRNA drugs: it had created siRNA that did not instantly break down in the bloodstream, got to

the cells it needed to, and entered them. That left the final hurdle of escaping the endosome. Without ionizable lipids to break the endosome membrane, how would naked siRNA ever make it out of endosomes and into the interior of the cell? The answer turned out to be unexpectedly simple. While most naked siRNA remains confined in the endosome, a small proportion slips out through some fluke occurrence, and more continues to be released over time. The proportion of siRNA leaking out of endosomes can be well under one percent, but often that is enough because siRNA is so potent – very little of it is actually needed to have its gene-silencing effect.



GalNAc delivers naked siRNA directly to liver cells

Three sugar molecules (GalNAc) attached to the siRNA are recognised by a receptor on liver cells, triggering uptake. Then the siRNA slowly leaks out of the endosome into the cell, while the receptor returns to the surface to be reused.



Adapted from Alnylam

The result is a massive difference in how long the drug lasts. Patisiran, which is carried by a lipid nanoparticle, needed to be dosed every three weeks. In contrast, Alnylam's newer drug, vutrisiran, a naked siRNA drug attached to a

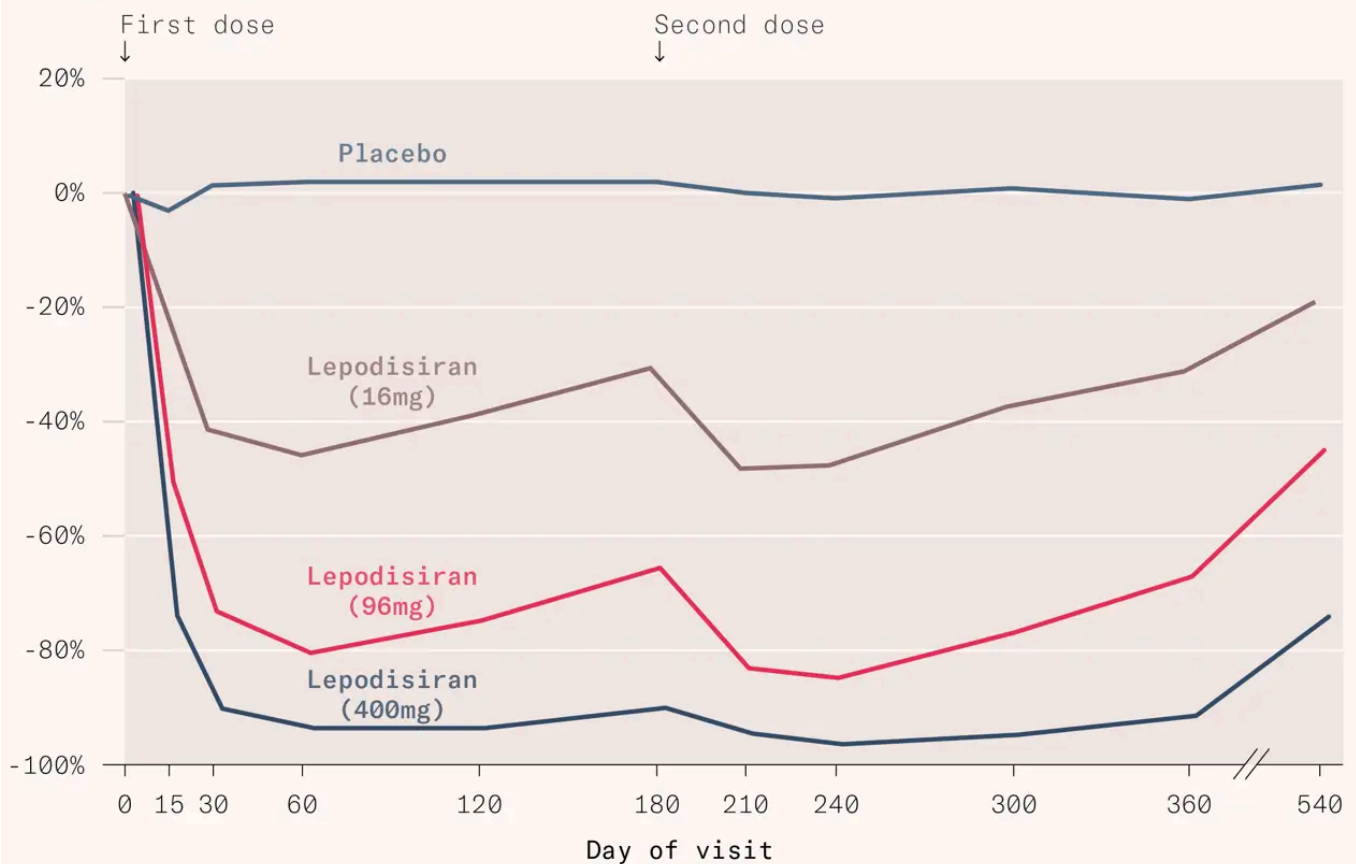
GalNAc sugar, is approved to treat the same medical condition, but needs to be dosed only once every three months. Other drugs based on naked siRNA and GalNAc now exist that only need to be topped up every six months. GalNAc siRNA drugs also have another advantage. Patisiran, with its lipid nanoparticle, carried the risk of triggering immune reactions, so patients had to be given immunosuppressants alongside it; but this precaution is unnecessary for those attached to GalNAc.



A new siRNA drug cuts lipoprotein(a), a harmful form of cholesterol, by up to 94%

Lipoprotein(a) is a form of LDL cholesterol that is largely genetic and cannot be reduced by diet or statins.

Change in lipoprotein(a) levels



Source: Nissen et al. (2025)

Simply put, temporarily switching off a precise gene in the liver is now a solved problem, and it can be achieved with drugs that require no more than four administrations per year. Multiple siRNA drugs have been approved for treating liver diseases, both rare and common, including treatments for hereditary transthyretin amyloidosis, hemophilia, high cholesterol, and acute hepatic porphyria, a nasty lifelong condition that causes attacks of pain, vomiting, and breathlessness because of a faulty enzyme. Many more siRNA drugs are in late-

stage trials, including treatments for high blood pressure, diabetes, and fatty liver disease.



Decades of science preceded a sudden burst of siRNA drugs

Foundational science

Liposomes discovered		1964
GalNAc liver receptor discovered		1968
First cationic lipid developed		1987
RNA interference found in petunias		1990
First ionizable lipid developed		1995
RNAi confirmed (Nobel Prize)		1998
siRNA silences gene in living animal		2006
Key ionizable lipid (MC3) developed		2012
Naked GalNAc-siRNA conjugates developed for delivery to the liver		2014

siRNA drug approvals

	2018		Patisiran (LNP-SiRNA) approved for hereditary transthyretin amyloidosis
	2019		Givosiran (GalNAc-siRNA) approved for acute hepatic porphyria siRNA
COVID mRNA vaccine deployed	2020		Lumasiran (GalNAc-siRNA) approved for primary hyperoxaluria type 1
	2021		Inclisiran (GalNAc-siRNA) approved for high cholesterol
	2022		Vutrisiran (GalNAc-siRNA) approved for hereditary transthyretin amyloidosis
	2023		Nedosiran (GalNAc-siRNA) approved for primary hyperoxaluria type 1
	2025		Fitusiran (GalNAc-siRNA) approved for haemophilia A/B Plozasiran (GalNAc-siRNA)



Beyond the liver

Every success story so far has involved the liver. But of course the liver is not the only organ in the body. The reason the liver dominates the story is not that siRNA will work only there but that the liver was an easy starting point. But once that approach worked, it suggested a broader idea – if siRNA can be stabilized and delivered by attaching a molecular ‘address’ to it, then in principle, different labels could be used to guide it to different organs.



That’s exactly what the field is now trying to do. By swapping out the sugar used to target the liver for other molecules, researchers are developing drugs targeted for other organs, including the kidneys, lungs, fat tissue, muscle, and brain. The competition is intense, and if any company stumbles, many other hungry startups, like Manifold Bio and Soufflé Therapeutics, will be hot on its tail.

Among these efforts, the highest-stakes trials are those aimed at the brain. They target one of modern medicine’s *bêtes noires*: Alzheimer’s disease. The most influential theory of Alzheimer’s, though it remains controversial, is that the disease is driven by a misfolded protein, amyloid beta, which is produced from a larger amyloid precursor protein. According to the theory, these misfolded proteins accumulate into clumps that trigger the accumulation of another protein, tau, which causes a loss of neurons and cognitive decline. This hypothesis naturally suggests two obvious intervention points: either reduce the production of amyloid precursor protein upstream or the production of tau downstream. Alnylam’s program targets the amyloid precursor, while Arrowhead’s targets tau.

These drugs will face different challenges than earlier siRNA drugs, which targeted the liver. The most immediate is accessing the brain, which is protected by the blood-brain barrier, a tight layer that seals and defends the brain from most molecules in the bloodstream. Alnylam’s current strategy is to bypass this barrier by injecting the siRNA directly into spinal fluid by ‘intrathecal injection’, where a needle inserted into the spine delivers drugs directly into spinal fluid, like in an epidural. Their siRNA is also attached to a fatty acid chain, which they believe will successfully deliver it to the relevant brain cells. But while this approach might prove effective, it’s also uncomfortable for patients and not ideal for frequent dosing.

Arrowhead is trying the bolder approach of delivering siRNA by injection under the skin, using a proprietary targeted siRNA molecule that's meant to cross the blood-brain barrier itself. There's a precedent for this approach, but no guarantee it will succeed.

Even if the amyloid hypothesis is correct and delivery succeeds, biology might still limit what siRNA can achieve. Famously, by the time Alzheimer's disease is diagnosed, the damage to the brain is often extensive enough that clearing amyloid plaques has little effect on slowing its progress. Whether targeting tau might be more effective at later stages remains an open question. Yet the broader point still stands: siRNA delivery beyond the liver is a matter of when, not if. If biologists can identify proteins that are critical to the progression of a disease, those diseases can become treatable or preventable.



The future of programmable medicines

siRNA falls into the category of therapies we could call 'programmable medicines'. Once medicinal chemists have worked out how to get a programmable medicine into a particular organ, all that's required to treat a new disease of that organ is to specify a new RNA or DNA sequence while holding the chemistry constant.^{8 9} So vutrisiran, the siRNA treatment for hereditary transthyretin amyloidosis, works according to the exact same principles as inclisiran, the siRNA that targets PCSK9 to reduce cholesterol levels.

siRNA is not the only programmable medicine in use, and other platforms have capabilities that siRNA lacks. For example, antisense oligonucleotides (ASOs) are short strands of synthetic genetic material that can alter RNA and be delivered similarly to siRNA. One such ASO, targeting the tau protein involved in Alzheimer's disease, has shown promising results in a phase two clinical trial. mRNA vaccines can trigger the production of certain proteins in our cells, and are a blindingly fast way to develop effective vaccines against viruses, as we saw during the Covid pandemic. CRISPR-based gene editing can permanently fix genes, rather than just temporarily switching them off, by directly altering cells' DNA.¹⁰ The common thread is that all these approaches link biological understanding directly to medicine. Once delivery is solved, the medicinal chemistry boot is off the biologist's face, and they can test new treatments simply by switching out the target genetic sequence.

Each of these forms of programmable medicine has already produced FDA-approved treatments. So far, only one class of problems has been solved by programmable medicine: gene silencing in the liver by siRNA. But sooner or later, the same will be true of other organs. As scientists solve each delivery problem, we'll experience an explosion of safe, convenient treatments against a myriad of diseases that were once considered untreatable. Soon, they will simply be waiting their turn.



1

Everyone interested in biomedical research should take some time to admire PROTACs, which destroy their targets instead of just temporarily blocking them, or anything else that Vertex Pharmaceuticals does.

2

These proteins can sometimes be drugged, as with daraxonrasib, which targets the 'undruggable' K-RAS protein, but doing so requires an immense amount of time, effort, and ingenuity.

3

Thalidomide (from the thalidomide scandal) was tested as a treatment for myeloma based on its ability to inhibit new blood vessel formation and hopefully starve tumors. It turned out to be effective, but for a completely different reason: it caused cells to break down the proteins that myeloma cells depend on. It does this by acting as a 'molecular glue' that links the proteins to another one that degrades it. A similar molecular glue approach is also used by daraxonrasib, Revolution Medicine's new drug for pancreatic cancer.

4

RNA's amazing versatility has led to the 'RNA world' hypothesis, that before life as we know it was a proto-life made of self-replicating RNA.

5

This calculation is based on the observation that Swedish patients with the condition have an annual risk of death about 2.64 times that of the general population. It assumes this higher relative risk applies throughout the lifespan and uses Swedish life table data from 2024 to estimate how many years of life are lost compared with average life expectancy: 9.17 years.

6

The 1,000-fold improvement in potency came from several design improvements. The FDA-approved lipid nanoparticles (LNPs) built around the ionizable lipid DLin-MC3-DMA have the optimal pKa (around 6.4), meaning they become positively charged at just the right acidity to destabilize endosomal membranes but not normal membranes. DLin-MC3-DMA also contains unsaturated fatty acids ('Lin' stands for linoleic acid) that give the molecule a slightly inverted cone shape, helping it further disrupt endosomal membranes and escape into the cell. Later efforts such as Moderna's SM-102 and Pfizer/BioNTech's ALC-0315 (used in the Covid-19 vaccines), or my own FO-32 and FO-35, push this inverted cone geometry even further and deliver cargo more efficiently.



7

Lipid nanoparticles were pivotal during the Covid-19 pandemic since they were the basis for delivering mRNA vaccines, and continue to be the leading nonviral platform for mRNA vaccines, gene therapy, and gene editing.

8

Or part of an organ, or particular cell type. For example, delivering genes to T cells in the body would have vast implications for autoimmune disease, fibrotic diseases, aging, and cancer via chimeric antigen receptor T cell therapy. This has already been achieved in animal models and various therapies are in clinical trials.

9

Many, likely most, clinical trials will still fail because the biological hypothesis of the disease is wrong. But I expect a very high success rate conditional on having the biology right, unlike with small molecule drugs, where a drug could fail even if the idea was correct.

10

Monoclonal antibody therapies could also arguably be classified as programmable medicine. I'd call them slightly intermediate between small molecule drugs and the nucleic acid-based programmable medicines discussed here.



Jacob Witten is a professor at Case Western Reserve University.



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